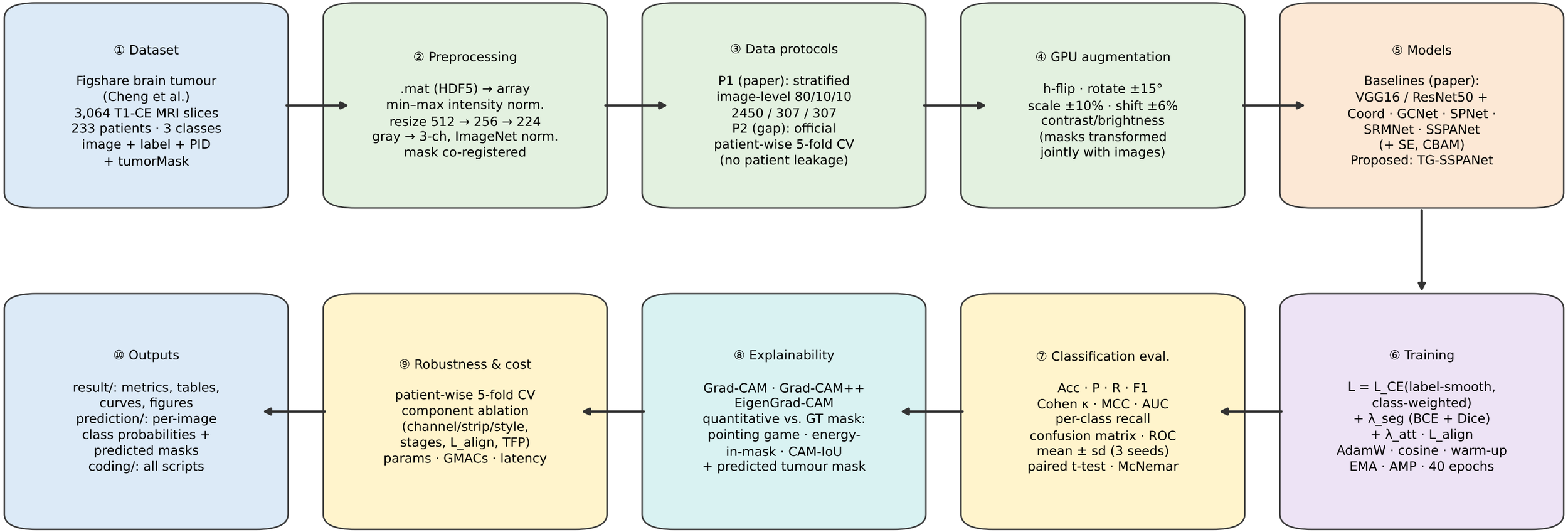
Experimental pipeline: paper reproduction + research-gap-driven TG-SSPANet



Research gaps addressed: G1 masks exist but were unused → mask-supervised attention + quantitative XAI · G2 image-level split leaks patients → patient-wise CV
G3 single-stage attention, no ablation → multi-stage TGSA + full ablation · G4 imbalance not handled → class-weighted, label-smoothed loss
G5 SE/CBAM never tested → added · G6 no measured cost / variance → params, GMACs, latency, mean ± sd, significance tests